Software Requirements Specification

for

Campus Connect

Version 1.0

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
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1. Introduction

1.1 Product Scope

In modern academic environments, effective communication and feedback play a crucial role in enhancing the quality of education and campus life. However, in many institutes, lecture feedback is often informal, unstructured, or collected through manual or fragmented systems. This makes it difficult for professors to analyse student responses, identify areas of improvement, and take data-driven decisions. Similarly, project opportunities are often shared through scattered channels, and there is no centralized platform for anonymous academic or campus-related discussions.

To address these challenges, the **Institute Academic Interaction & Feedback Platform** is proposed as a comprehensive, web-based solution designed exclusively for students and professors of an institute. The platform aims to streamline academic feedback collection, provide meaningful insights to faculty, centralize project listings, and enable anonymous campus-wide interaction in a secure and structured manner.

The system introduces a **course-based feedback mechanism**, where students can provide structured feedback for each lecture using a standardized rating scale. Alongside numerical feedback, students are encouraged to share anonymous comments and ask lecture-related questions, fostering open and honest academic communication. Professors, in turn, gain access to visual analytics and insights that summarize student feedback, helping them evaluate teaching effectiveness and improve instructional quality.

In addition to feedback management, the platform includes a **project module** that allows professors to post academic or research projects with relevant details, enabling students to discover opportunities in a centralized and transparent manner. **Students can also apply for these projects directly through the platform**, streamlining the project selection process and reducing dependency on external communication channels. Furthermore, a **campus overview module** facilitates anonymous questions and discussions related to campus life, academics, and general concerns, encouraging inclusive participation from both students and professors.

By digitizing and integrating these academic workflows into a single platform, the system reduces communication gaps, enhances transparency, promotes student engagement, and supports continuous improvement in teaching and learning processes. Ultimately, the platform aims to strengthen the academic ecosystem by fostering data-driven insights, collaboration, and open dialogue within the institute.

1.2 Intended Audience and Document Overview

1.2.1 Intended Audience

This Software Requirements Specification (SRS) document is intended for the following stakeholders involved in the development and use of the Institute Academic Interaction & Feedback Platform:

- **Developers**
  Developers will use this document to understand the system's functional and non-functional requirements. It provides clear descriptions of system modules, workflows, data handling, and access control mechanisms required for effective implementation.

- **Project Managers**
  Project managers will rely on this document to monitor progress, manage resources, and ensure that development aligns with project objectives. The document offers a structured overview of the system scope, features, and constraints.
- **Institute Authorities / Academic Administrators**
  Academic administrators can use this document to understand how the system improves feedback collection, teaching quality assessment, and academic coordination within the institute.
- **End Users (Students and Professors)**
  Students and professors will benefit from the descriptions of system functionality and workflows, helping them understand how the platform supports feedback submission, project discovery, and campus-level interaction.
- **Testers**
  Testers will use this document to derive test cases, verify system behaviour, and ensure that the platform meets its specified requirements and quality standards.
- **Documentation Writers**
  Writers preparing user manuals, help guides, or training material will use this SRS as a foundational reference for system features, terminology, and usage flows.

1.2.2 Document Overview

This SRS document is organized into the following sections:

1. **Introduction**
   Provides an overview of the system, defines its scope and objectives, identifies the intended audience, and outlines the structure of the document. It also includes definitions, acronyms, document conventions, and references.

2. **Overall Description**
   Presents a high-level description of the system, including product perspective, user characteristics, operating environment, constraints, assumptions, and dependencies.

3. **Requirements**
   Specifies the detailed requirements of the system, including:
   - **Functional Requirements**: Core system features such as lecture feedback, analytics, project management, and campus discussions.
   - **Non-Functional Requirements**: Performance, security, scalability, usability, and reliability constraints.
   - **External Interface Requirements**: User interfaces, software interfaces, and communication mechanisms.
   - **Use Case Model**: Scenarios and workflows illustrating system behaviour.

4. **Additional Requirements**
   Includes supplementary information such as data dictionaries, validation rules, and other requirements not covered in earlier sections.

5. **Appendices**
   Contains supporting materials such as detailed diagrams, group logs, and references used during the project lifecycle.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
CSS	Cascading Style Sheets
HTML	Hyper Text Markup Language
OTP	One Time Password
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience
RBAC	Role-Based Access Control

1.4 Document Conventions

Formatting Conventions

1. The document is written in Arial font, size 11, with 1.15 line spacing and 1-inch margins.
2. Main headings use Arial font, bold, size 18.
3. Subheadings use Arial font, bold, size 14.
4. Sub-subheadings use Arial font, bold, size 13.
5. Bullet points are used for structured listing.
6. Important terms are highlighted in bold where necessary.

Naming Conventions

1. **User** – Any authenticated individual using the system.
2. **Student** – An enrolled institute student.
3. **Professor** – An institute faculty member.
4. **Course** – An academic subject offered by the institute.
5. **Lecture** – A single class session associated with a course.
6. **Feedback** – Structured rating-based evaluation submitted by students.
7. **Comment** – Anonymous textual feedback.
8. **Question** – Lecture-related or campus-related query posted by users.

1.5 References and Acknowledgments

1. Lucidchart was used for creating system diagrams, flowcharts, and ER models.
2. Canva was used for graphical representation of the system overview and module summaries.
3. Existing academic feedback systems and learning management platforms were studied to understand best practices and usability considerations.
4. The project team acknowledges the guidance and feedback provided by faculty mentors and peers during the conceptualization and design phases.

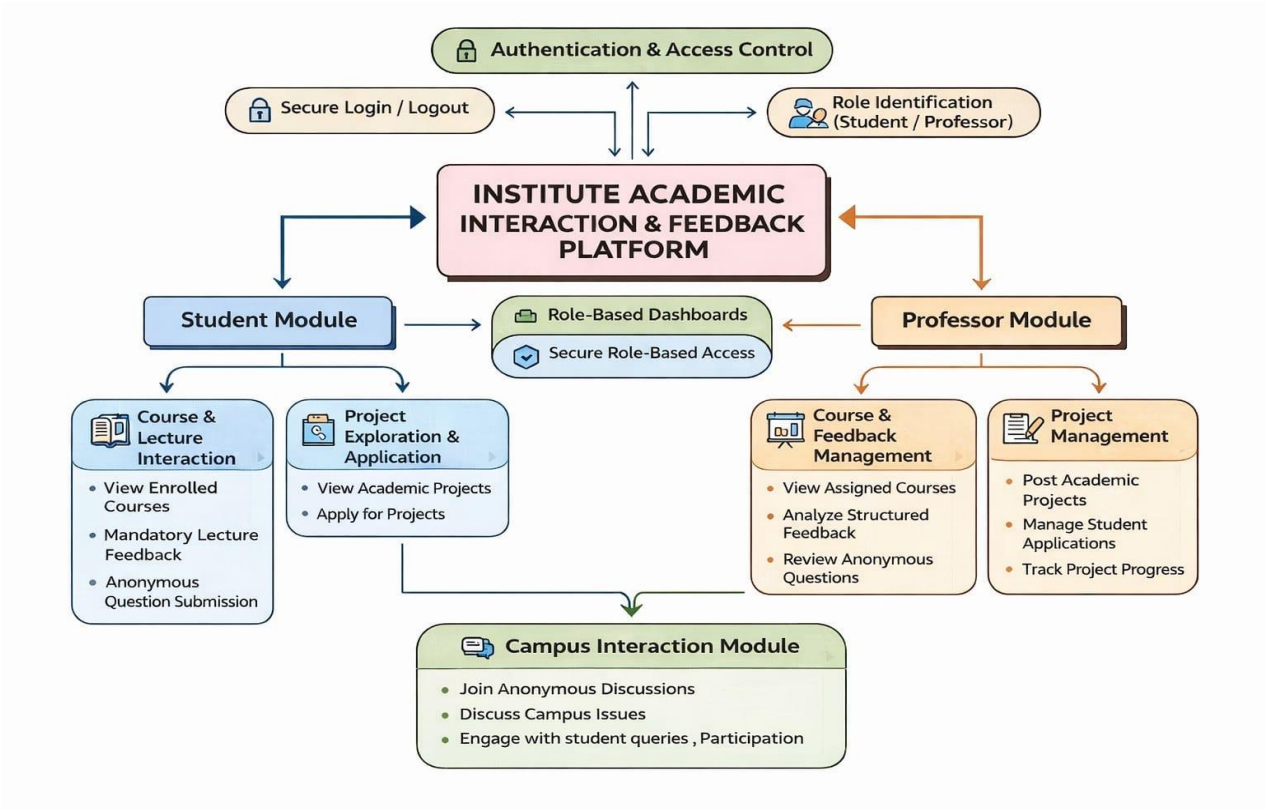
2. Overall Description

2.1 Product Overview

The **Institute Academic Interaction & Feedback Platform** is a web-based software solution designed to enhance academic communication, feedback collection, and project collaboration within an educational institute. The platform addresses the absence of a structured lecture feedback system, difficulty in feedback analysis, lack of centralized project listings, and limited anonymous campus interaction.

The system provides a centralized digital environment where students and professors can interact efficiently through role-based dashboards. It enables students to submit structured feedback for each lecture, participate in anonymous academic discussions, and explore available academic projects. Professors can analyze lecture feedback using visual analytics, manage academic projects, and engage with student queries. By integrating feedback, project management, and campus-wide interaction into a single platform, the system aims to improve teaching quality, student engagement, and overall academic transparency.

The platform operates as a secure, institute-only system with role-based access control, ensuring that users interact only with features relevant to their academic role. It is designed to be scalable, user-friendly, and suitable for real-time academic environments.



2.2 Product Functionality

The platform functions as an academic management and interaction system with the following core functionalities:

2.2.1 Authentication and Access Control

- Institute-only user registration
- Role identification during signup (Student or Professor)
- Secure login and logout functionality
- Role-based dashboards and access control

2.2.2 Student Module Functionality

The Student Module provides the following core functionalities:

a) Course and Lecture Interaction

- View list of enrolled courses
- Calendar-based lecture view where each date represents a lecture
- Mandatory structured feedback submission for every lecture
- Feedback parameters include clarity, pace, engagement, understanding, and overall effectiveness
- Optional anonymous comments for additional opinions
- Anonymous question submission related to lectures

b) Project Exploration and Application

- View academic projects posted by professors
- Access detailed project descriptions and requirements
- Apply for projects directly through the platform
- Track projects applied for by the student

c) Campus Interaction

- Participate in campus-wide anonymous discussions
- Post academic questions and view announcements
- Engage with peers and professors while maintaining anonymity

2.2.3 Professor Module Functionality

a) Course and Feedback Management

- View assigned courses and lectures
- Access lecture-wise structured feedback
- Graphical visualization of feedback trends and ratings
- Analyze parameters such as clarity, engagement, and understanding
- View anonymous student remarks and questions

b) Project Management

- Create and post academic projects with descriptions and requirements
- View list of projects offered by the professor
- View and manage student applications for projects
- Track project status and student interest

c) Campus Interaction

- View campus-wide anonymous posts and discussions
- Post academic announcements
- Engage with student queries in a controlled environment

2.3 Design and Implementation Constraints

1. The system must efficiently store and process large volumes of feedback data, user interactions, and project information.
2. Real-time data updates are required for feedback analytics, campus discussions, and dashboards.
3. The application must support concurrent access by multiple users during peak academic hours.
4. The user interface must be simple, intuitive, and accessible to users with varying levels of technical expertise.
5. Data privacy and anonymity must be maintained, especially for feedback and campus discussions.

2.4 Assumptions and Dependencies

Assumptions

1. All users belong to the same institute and have valid institutional credentials.
2. Users have access to a stable internet connection.
3. Students are required to submit feedback for lectures before accessing subsequent course content.
4. Professors regularly review feedback to improve teaching effectiveness.

Dependencies

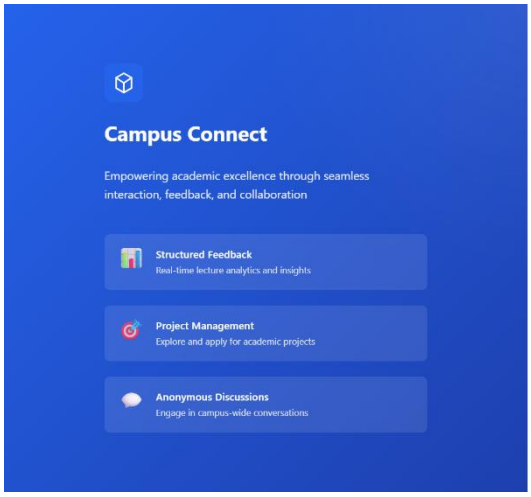
1. The front-end depends on HTML, CSS, and JavaScript for creating an interactive and responsive user interface.
2. The system relies on modern JavaScript libraries for data visualization and user interaction.
3. The backend is dependent on a web framework (such as Django) and a relational database for secure data storage and processing.
4. Graph and analytics libraries are required for visual representation of feedback data.

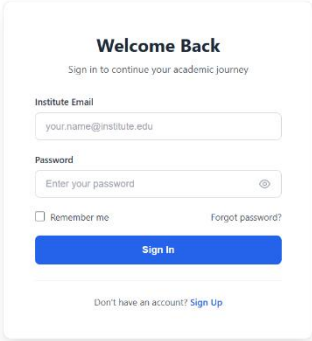
3. Specific Requirements

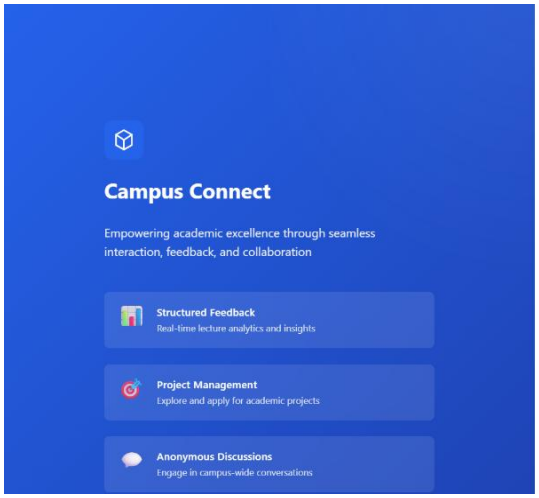
3.1 External Interface Requirements

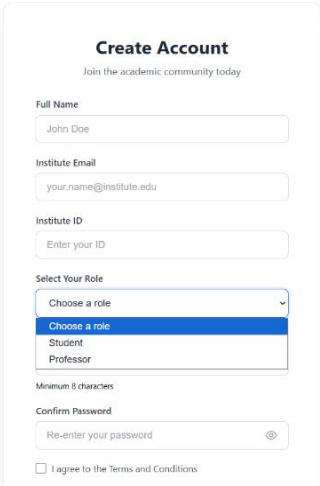
3.1.1 User Interfaces

1. Login/ Sign Up Page









1.1 Login

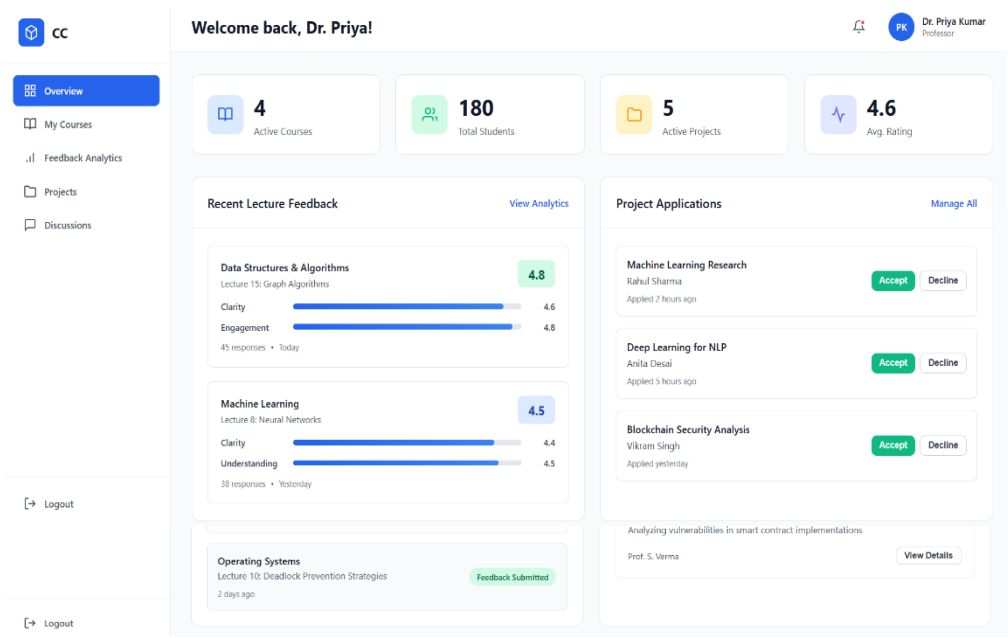
- The login page prompts users to enter their registered institute email and password.
- Upon successful authentication, users are redirected to the dashboard/home Spage.
- "Forgot Password" option available for password recovery via OTP verification.
- Error messages displayed for invalid credentials
- Session management to keep users logged in

1.2 Signup

- New users register using their institute email with role selection (Student or Professor)

- **OTP Verification:** System sends OTP to provided email with 10-minute expiration; users enter OTP to verify email ownership
- **Password Creation:** Users set secure password (minimum 8 characters) with confirmation field and strength indicator to prevent errors
- After successful registration, users are redirected to role-specific dashboard.

2. Dashboard (Home Page)



2.1 Student Dashboard

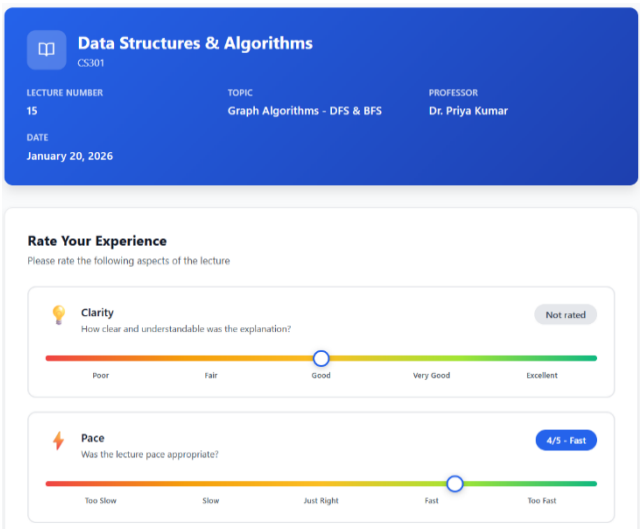
- **Quick Stats Display:** Shows total courses enrolled, pending feedback count, active project applications, and discussion participation metrics.
- **Pending Feedback Management:** Displays courses requiring feedback with course details, instructor name, last lecture date, and direct "Give Feedback" action button.
- **Recent Activity Timeline:** Tracks and displays latest submissions including feedback forms, project applications, and forum posts with timestamps and status.
- One-click access to submit feedback, browse projects, view courses, and create discussion posts.

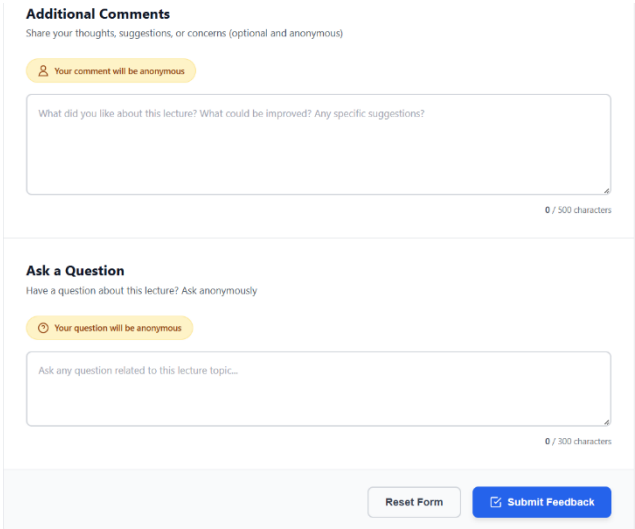
2.2 Professor Dashboard

- **Analytics Overview:** Displays total courses taught, total feedback received with percentage trends, active projects posted, and student queries answered.
- **Feedback Analytics:** Shows average ratings across five parameters (Content Quality, Delivery, Clarity, Engagement, Pace) with trend indicators and comparison charts over time periods.
- **Recent Feedback Review**: Table view of latest feedback submissions with course name, date, average rating, sentiment analysis (positive/neutral/negative), and detailed view option.

- **Project Application Management:** Displays active projects with application counts, recent applicant profiles with skills, and quick action buttons (Accept/Review/Decline applications).
- **Student Query Response:** Shows recent discussion forum questions from students with reply functionality and "mark as answered" option for closure.

3. Lecture Feedback





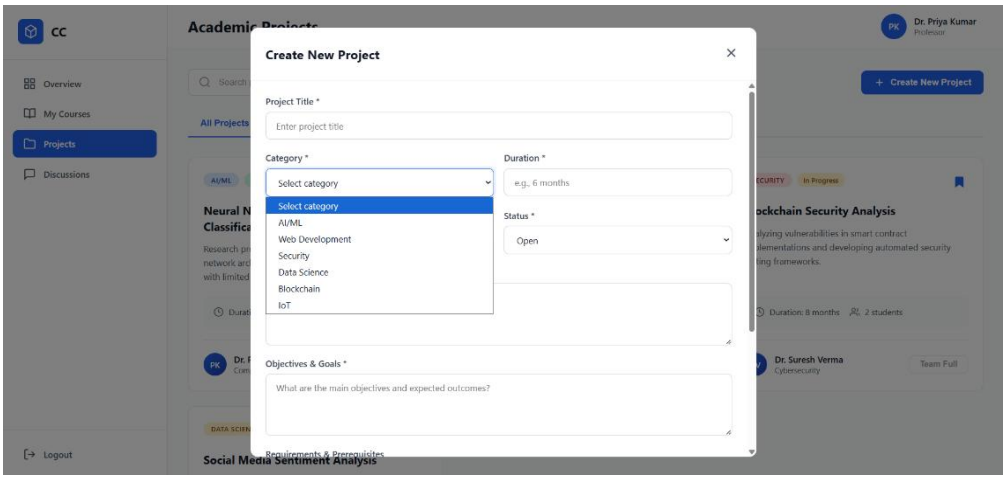
3.1 Student View

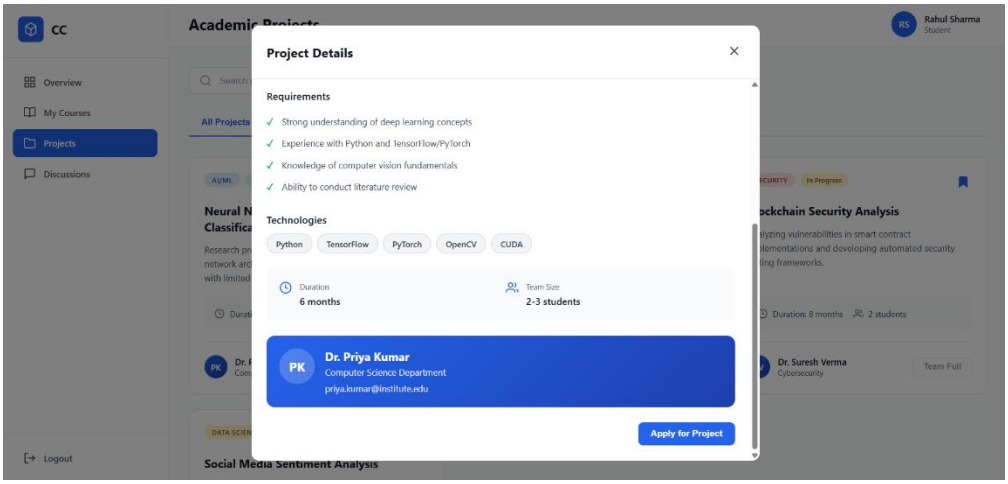
- **Rating Parameters:** Five-point scale (1-5 stars) for rating Content Quality, Teaching Delivery, Clarity of Explanation, Student Engagement, and Lecture Pace with visual star indicators.
- **Anonymous Comments:** Optional text area for detailed feedback with 500-character limit, submitted anonymously to encourage honest responses.
- **Draft Saving:** Auto-save functionality to preserve incomplete feedback forms with manual save option to complete later.
- **Submission Validation:** Ensures all mandatory rating fields are filled before submission with error highlighting for missing fields.
- **Confirmation & Redirect**: Success message upon submission with automatic redirect to dashboard and removal from pending feedback list.

3.2 Professor View

- **Aggregated Analytics:** View consolidated feedback statistics for each course including average ratings per parameter and total responses received.
- **Trend Analysis:** Graphical representation of rating trends over multiple lectures/semesters with comparison capabilities.
- **Anonymous Comment Review:** Read all student comments without identifying individual students to maintain confidentiality and academic freedom.

4. Academic Projects





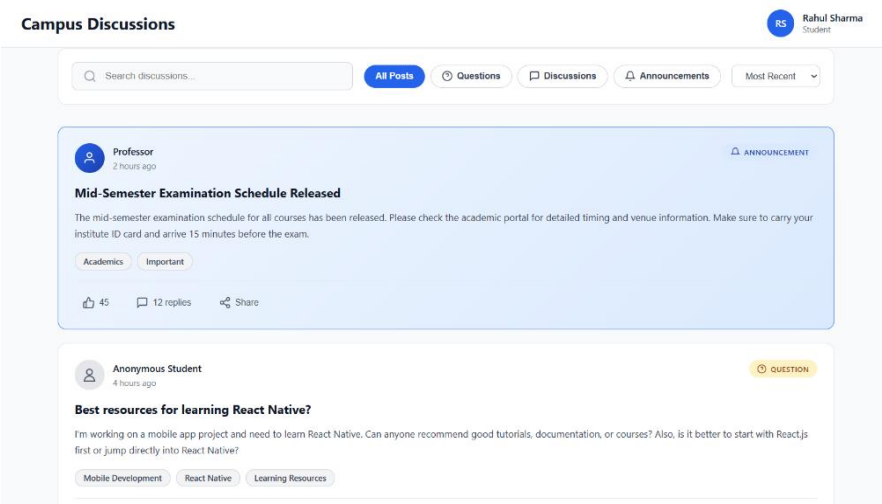
4.1 Student Features

- **Project Discovery:** Browse all available academic projects with search by keywords (title/description) and filter by category.
- **Project Details View:** Modal displaying complete project information including overview, objectives, requirements checklist, technology stack, duration, team size, and professor contact details.
- **Application Submission:** Apply to projects with motivation statement, relevant skills/experience description, and availability information with character limits.
- **Application Tracking:** "My Applications" tab shows all submitted applications with status (Pending/Accepted/Rejected) and submission timestamps.

4.2 Professor Features

- **Project Creation:** Comprehensive form to create new projects with title, category, duration, team size requirements, status selection, detailed description, objectives, prerequisites, required technologies, and skills.
- **Project Management:** "My Projects" tab displays all created projects with application counts, status management (Open/Closed), and edit/delete options.
- **Application Review:** View student applications with their profiles, motivation, skills, and availability with quick action buttons (Accept/Reject/Shortlist).
- **Student Communication:** Direct access to applicant profiles for contacting shortlisted candidates through platform messaging.

5. Campus Discussion Forum



- **Anonymous Posting System:** All student posts and replies are completely anonymous to encourage open academic discussion and honest opinions.
- **Post Type Selection:** Create three types of posts - Questions (students seeking help), Discussions (general academic topics), and Announcements (professors only for important updates).
- **Post Creation:** Title input (150-char limit), rich text content area (2000-char limit), optional tag addition (up to 5 comma-separated tags), and character counters with real-time validation.
- **Draft Management:** Save posts as drafts to complete later with auto-save functionality for accidental closures.
- **Anonymous Replies:** All replies maintain anonymity.

3.1.2 Hardware Interfaces

The user does need any hardware interfaces.

3.1.3 Software Interfaces

The user does not need any special software. Only a web browser and a stable internet connection are needed.

1.1 Functional Requirements

3.2.1 F1: Sign Up / Sign In

The system should provide authentication functionality for two user roles: **Student** and **Professor**. Users shall register and log in using their **email ID as a unique identifier**.

- A registered user shall be able to **sign in using email ID and password**.
- A new user shall be able to sign up as a Student or Professor with **OTP-based email verification**.

- The system should provide a **'Forgot Password?'** option with password reset through **OTP verification**.

Student Sign Up:
A student shall register using an **anonymous username**, email ID, and password.

Professor Sign Up:
A professor shall register using their **name, email ID, and password**. The system shall verify the professor's email ID against the admin panel's professor database to ensure authorized access.

3.2.2 F2: User Dashboard

- The dashboard shall display the list of **courses enrolled by the student** or **courses taught by the professor** for the current semester.
- For each course, the dashboard shall display the **course name, course duration** (start date and end date)**,** and **professor's name**.
- For each course, the dashboard shall display the **corresponding class sessions**, showing the **day and date**.
- The dashboard shall display a **visual indicator showing the number of unread comments or replies** for each course and class session.
- The dashboard shall allow users to navigate to the class discussion page.

This dashboard provides a centralized view of courses, class schedules, and discussion activity, enabling users to easily track recent classes.

3.2.3 F3: Course Creation and Scheduling (Professor)

- The system shall allow the professor to **create a course** by providing the course name.
- The system shall allow the professor to select the **course start date and end date using a calendar-based date selection feature**.
- The system shall allow the professor to select the **days of the week** on which the class is conducted.
- Based on the selected days and course duration, the system shall automatically generate **all class sessions for the course** at the time of course creation.
- The system shall allow the professor to **add one or more co-instructors** to a course by providing their **email IDs**.
- The system shall validate each entered email ID while adding a co-instructor. If the email ID **does not exist in the system** or is **already associated with the course**, the system shall display an **appropriate error message**.
  If the email ID is **valid and not already associated**, the system shall **directly add the professor as a co-instructor**.
- Added co-instructors shall be granted the same course access and privileges as the **primary professor**.
- The system shall allow the professor to **delete an existing course**.

This feature enables efficient course creation, scheduling, and collaboration among instructors by automatically generating all class sessions upfront and providing controlled access to multiple professors involved in the course.

3.2.4 F4: Student Enrolment via Excel Upload and Manual Addition

- The system shall allow the professor to **upload an Excel file** to add students to a course. If the uploaded file is **not a valid Excel file**, the system shall display an **appropriate error message**.
- The system shall extract student email IDs from a column named '**Email'** in the uploaded Excel file. If the '**Email' column is not present**, the system shall display an **error message indicating the missing column** and cancel the process.
- The system shall **validate each extracted email ID**. If an email ID is **invalid** or **already exists,** the system shall **skip the entry**.
- After processing the Excel file, the system shall display a **summary** showing the **number of students successfully added** and the **number of entries skipped** (if any) and their **corresponding email ID.**
- The system shall allow the professor to **manually add a student** to the course by entering the **student's email ID**. If the entered email ID is **invalid or already exists**, the system shall display an **appropriate error message**.
- The system shall allow the professor to **view the list of enrolled students** for the course.
- The system shall provide a **search option** to search enrolled students using their **email ID**.

This feature ensures accurate, duplicate-free student enrolment while providing clear validation and error feedback for both Excel-based and manual enrolment actions.

3.2.5 F5: Anonymous Discussion and Replies

- The system shall allow students to **post comments and reply anonymously** under a specific class session.
- Each comment posted by a student shall display the **anonymous username** instead of the real identity.
- The system shall allow professors to **post comments and replies** under class sessions.
- The system shall visually distinguish between:
    - **Messages posted by the current user**, which shall be **aligned to the right**.
    - **Messages posted by other users**, which shall be **aligned to the left**.
- The system shall allow users to **reply to an existing comment**, creating a threaded or contextual reply under the original comment.
- Replies shall clearly indicate the **comment being replied to**, while maintaining anonymity for student users.
- The system shall display comments and replies in **chronological order**, ensuring smooth conversation flow.
- The system shall update the discussion view in near real-time so that **new comments and replies are visible without requiring a page refresh**.

- The system shall mark comments and replies as **unread** until the user views the discussion.
- The system shall require students to submit **compulsory feedback** for each class session using a **multiple-choice question (MCQ) format** to rate the class.
- The system shall associate the submitted feedback with the corresponding class session for analytics and evaluation purposes.

This feature enables structured, anonymous, and interactive discussions within each class session, ensuring clarity in conversations while preserving student anonymity and improving user experience through clear visual distinction.

3.2.6 F6: Project Listing, Application Review, and Management (Professor)

- The system shall allow professors to **create and publish project opportunities** by providing a project title and description.
- The system shall allow professors to define **eligibility criteria** for a project based on batch, branch, and minimum CPI.
- The system shall allow professors to mark any eligibility criterion as **open to all**, indicating no restriction for that criterion.
- The system shall allow professors to **edit or remove** an existing project listing.
- The system shall allow professors to **view all applications** received for a project.
- The system shall display complete application details to the professor, including the student's **name, roll number, branch, batch, CPI, and uploaded resume**.
- The system shall allow professors to **sort and filter applications** based on branch, batch, and CPI.
- The system shall allow professors to **accept or reject** student applications.
- The system shall record the decision taken for each application and associate it with the respective student and project.

This feature enables professors to advertise projects, define flexible eligibility criteria, and efficiently review and manage student applications.

3.2.7 F7: Project Discovery, Application, and Status Tracking (Student)

- The system shall allow students to **view the list of available projects**.
- The system shall display complete project details, including the project description and eligibility criteria.
- The system shall allow students to **search and filter projects** based on branch, batch, and CPI.
- The system shall allow students to **apply for a selected project**.
- While applying for a project, the system shall require the student to provide **name, roll number, branch, CPI, and an uploaded resume**.
- The system shall validate the submitted information and resume and display an **appropriate error message** if any required detail is missing or invalid.

- Upon successful submission, the system shall **store the application** and link it to the respective project.
- The system shall allow students to **view the status of their applications**.
- The application status shall be displayed as **pending, accepted, or rejected**, and any update made by the professor shall be **reflected in the student's account**.

This feature allows students to discover relevant projects, submit structured applications, and track application outcomes transparently.

3.2.8 F8: Course Analytics and Feedback Visualization

- The system shall allow professors to **view analytics for a selected course**.
- The system shall process **student feedback data** associated with the selected course.
- The system shall present feedback analytics to the professor using **graphical visualizations**, including **bar charts** and **trend lines**.
- Bar charts shall be used to represent **categorical feedback metrics**.
- Trend lines shall be used to represent **feedback trends over time**.
- The system shall display the analytics graphs on the **professor's dashboard**.
- If no feedback data exists for the selected course, the system shall display an **appropriate notification indicating that no data is available**, instead of showing empty graphs.

This feature enables professors to analyze compulsory student feedback effectively by visualizing feedback patterns and trends, supporting data-driven evaluation of course performance.

3.2.9 F9: Account Management

- The system shall provide users with access to **account management features** through the dashboard.
- The system shall provide a **Profile Settings** option that allows users to update permitted account details.
  Students shall be able to modify their **anonymous username**, while professors shall be able to update their **name**.
- The system shall provide a **Change Password** option that allows users to securely update their account password using the **existing password**.
- The system shall provide a **Logout** option that terminates the user session securely and prevents unauthorized access.

This feature enables users to manage their account information and security settings while ensuring safe session handling.

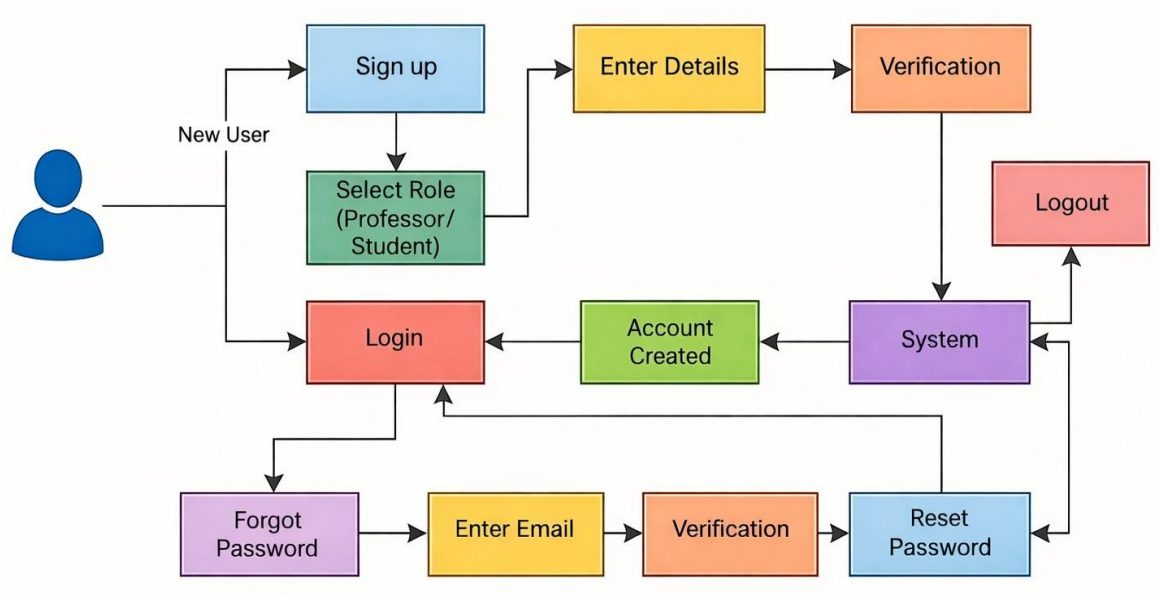
3.2.10 F10: Admin Panel

- The system shall provide an **admin panel** accessible only to authorized administrators.
- The admin panel shall allow administrators to **add, remove, and manage professor email IDs**.
- The admin panel shall allow administrators to **control professor authorization** within the system.
- Only **email IDs present in the admin panel** shall be allowed to **register as professors**.

This feature ensures system integrity by restricting professor privileges to authorized users and preventing unauthorized access or misuse.

3.2 Use Case Model

3.2.1 Use Case #1 (Signup/Login)



Author - Team DevAxis

Purpose - To register new users (Students/Professors) with role identification and authenticate them to access the system. It also handles password recovery.

Requirements Traceability - Institute-only user registration, Role identification (Student/Professor), Secure login/logout.

Priority - High

Preconditions -

For Sign Up: User must have a valid institute email address.

For Login: User must have a registered account.

Post conditions -

Sign Up: A new user account is created with a specific role (Student or Professor).

Login: User is authenticated and redirected to their role-specific dashboard.

Actors - New User, Registered User (Student/Professor), System.

Exceptions -

Invalid Credentials: System displays an error message if email/password is incorrect.

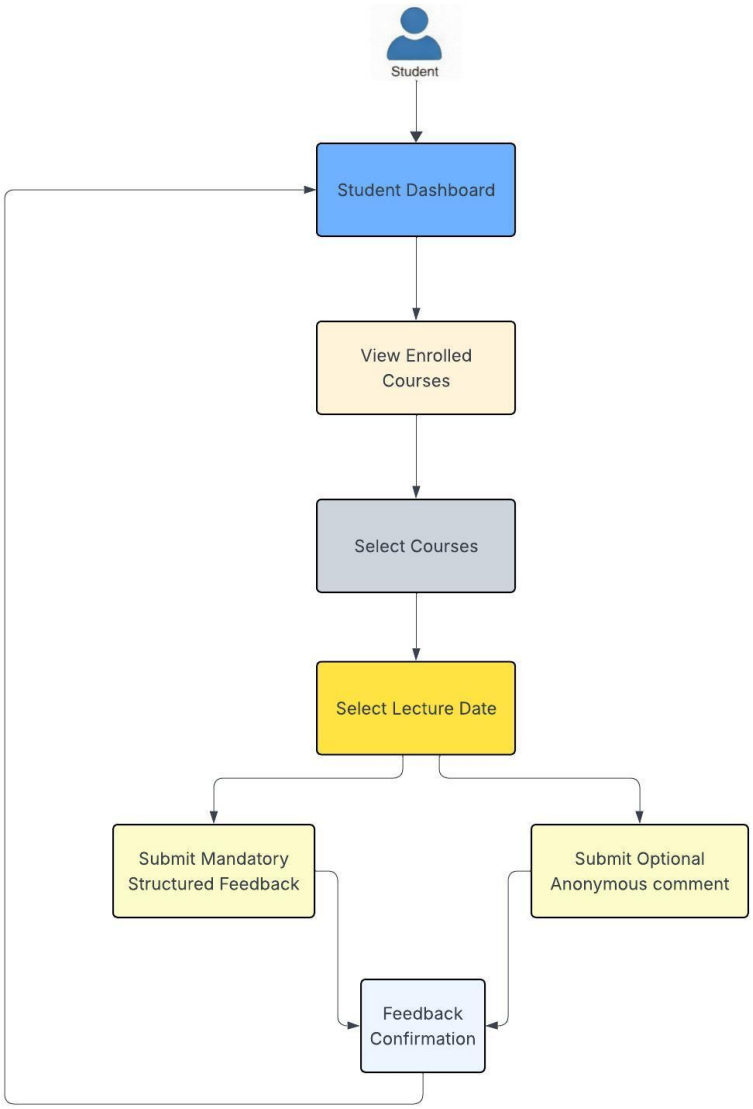
User Already Exists: System prevents duplicate registration with the same email.

Verification Failed: OTP/Email verification fails or times out.

Includes - None.

Notes/Issues - The flowchart explicitly includes a "Select Role" step during sign-up, ensuring users are correctly bucketed as Student or Professor immediately.

3.2.2 Use Case #2 (Submit Lecture Feedback)



Author - Team DevAxis

Purpose - To allow students to provide mandatory structured feedback and optional comments for a specific lecture they attended.

Requirements Traceability - Mandatory structured feedback submission, Calendar-based lecture view, Optional anonymous comments.

Priority - High

Preconditions –

Student is logged in.

Student is enrolled in the specific course.

Post conditions -

Feedback is successfully saved to the database.

Professor's analytics are updated (asynchronously).

Student receives a "Feedback Confirmation".

Actors - Student, System.

Exceptions -

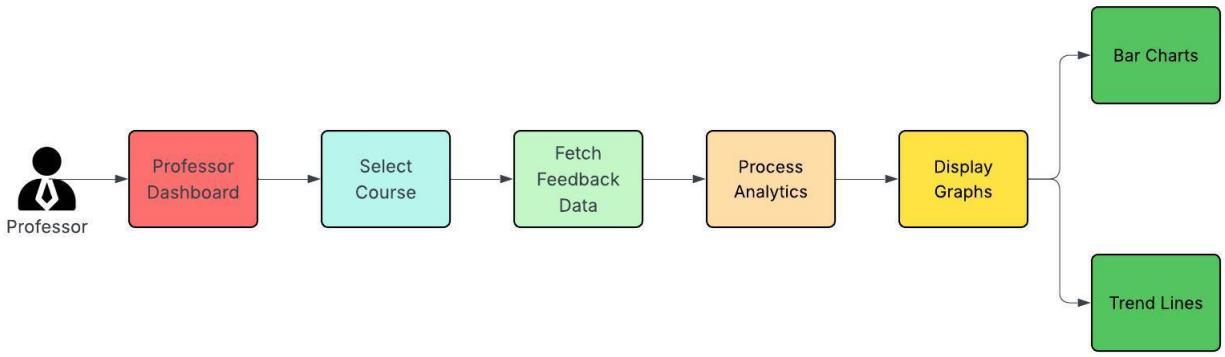
Missing Mandatory Fields: System prompts user to complete structured feedback before submission.

Network Error: Submission fails; user is asked to retry.

Includes - None.

Notes/Issues - The flow confirms that structured feedback is mandatory while the anonymous comment section is optional.

3.2.3 Use Case #3 (View Course Analytics)



Author – Team DevAxis

Purpose - To process student feedback data and present it to the professor in visual formats (graphs and charts) for analysis.

Requirements Traceability - Visual analytics for professors, Graphical visualization of feedback trends (Bar Charts, Trend Lines).

Priority - High

Preconditions -

Professor is logged in.

Feedback data exists for the selected course.

Post conditions -

Analytics graphs (Bar Charts, Trend Lines) are displayed on the dashboard.

Actors - Professor, System.

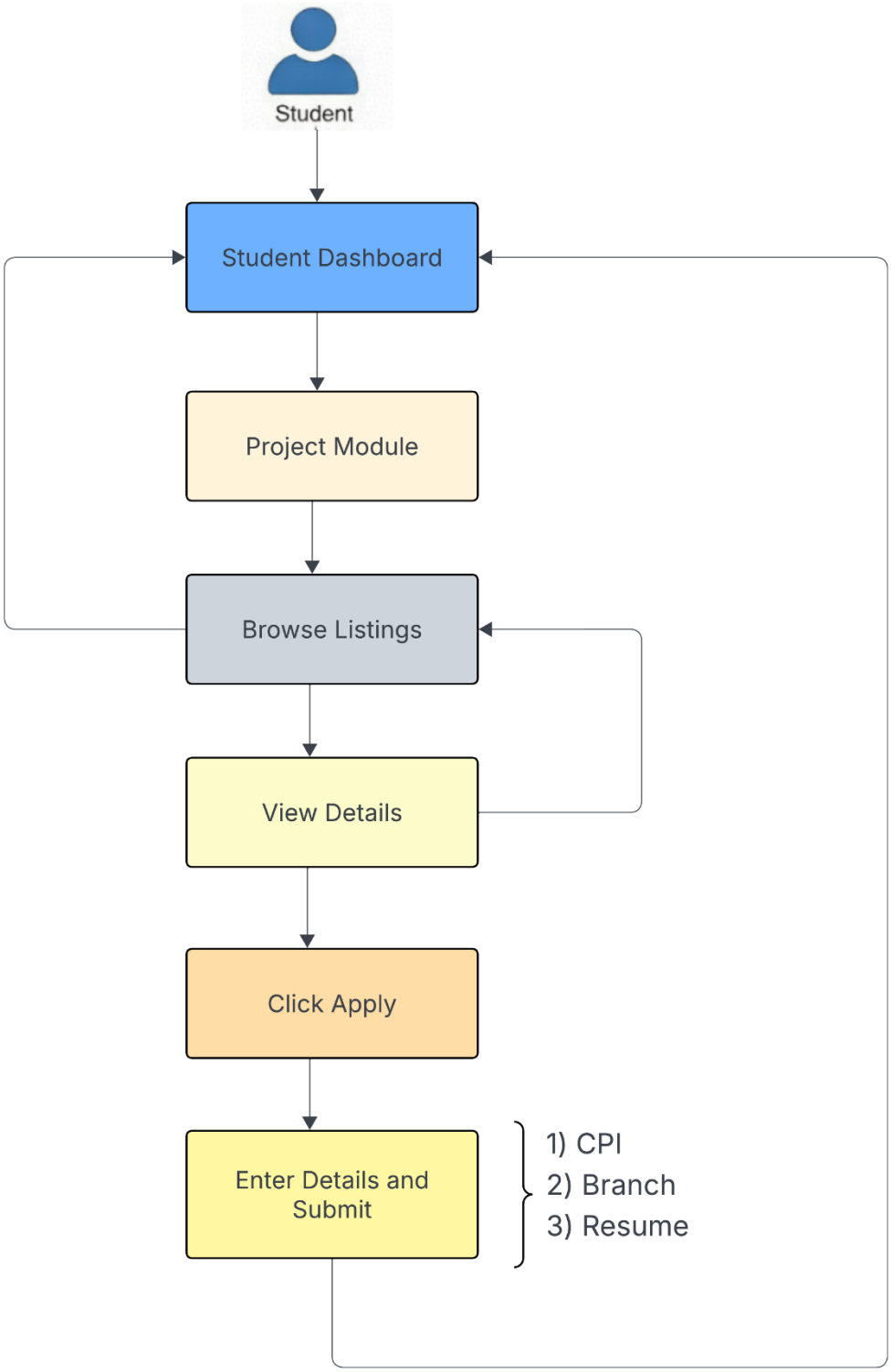
Exceptions -

No Data Available: If a course has no feedback yet, the system should display a notification instead of empty graphs.

Includes - None.

Notes/Issues - The flowchart specifies two types of visualization: "Bar Charts" (likely for categorical ratings like Clarity) and "Trend Lines" (likely for performance over time).

3.2.4 Use Case #4 (Student Project Module)



Author - Team DevAxis

Purpose - To allow students to browse project listings, view details, and submit an application with their academic credentials.

Requirements Traceability - View academic projects, apply for projects directly, Enter details (CPI, Branch, Resume).

Priority - Medium

Preconditions -

Student is logged in.

Projects are available in the system.

Post conditions -

Application is submitted and appears in the Professor's "View Applications" list.

Actors - Student, System.

Exceptions -

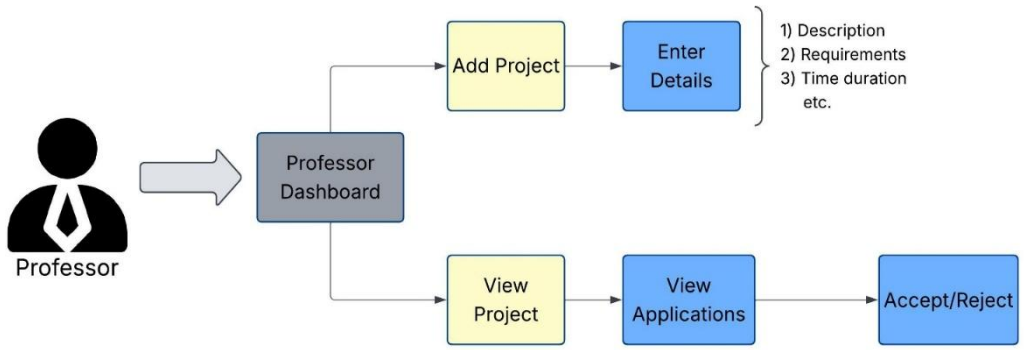
Missing Documents: User fails to upload the required Resume.

Application Closed: Student attempts to apply for a project that is no longer accepting applicants.

Includes - None.

Notes/Issues - The flowchart confirms specific data points for the application form: CPI, Branch, and Resume upload.

3.2.5 Use Case #5 (Professor Project Module)



Author - Team DevAxis

Purpose - To enable professors to post new projects with specific requirements and manage incoming student applications (Accept/Reject).

Requirements Traceability - Create and post academic projects, Project descriptions and requirements, View and manage student applications.

Priority - Medium

Preconditions -

Professor is logged in.

Post conditions -

Add Project: A new project is visible in the "Project Module" for students.

Manage Applications: Student application status is updated to Accepted or Rejected

Actors - Professor, System.

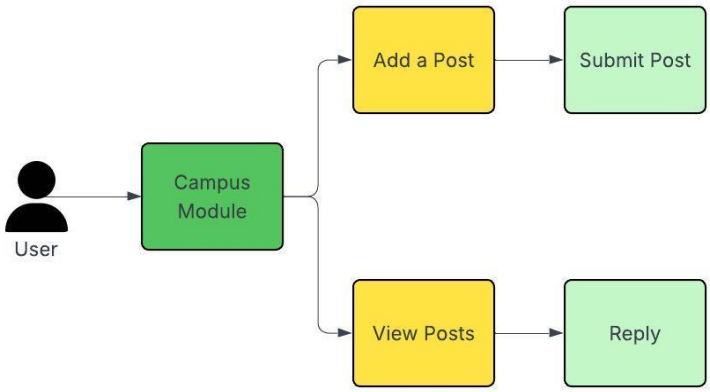
Exceptions -

Incomplete Details: Professor attempts to post a project without a description or title.

Includes - None.

Notes/Issues - The input details explicitly include "Time duration," which is a specific requirement for the project posting form.

3.2.6 Use Case #6 (Campus Interaction Module)



Author - Team DevAxis

Purpose - To facilitate campus-wide communication through posts and replies (likely anonymous or public).

Requirements Traceability - Campus-wide anonymous discussions, Post questions and announcements, Engage with peers.

Priority - Low

Preconditions -

User (Student or Professor) is logged in.

Post conditions -

Post or Reply is published to the Campus Module feed.

Actors - User (Student/Professor), System.

Exceptions -

Content Policy Violation: If content filtering is active, inappropriate posts may be blocked.

Includes - None.

Notes/Issues - The flow is simple (Add/View/Reply). It does not explicitly show an "Anonymity Toggle," so this might need to be handled within the "Add a Post" screen

4. Other Non-functional Requirements

4.1 Performance Requirements

- **System Availability**
  The system shall maintain a minimum uptime of **99.5%**, excluding scheduled maintenance windows, to ensure uninterrupted access to feedback services, project listings, and campus interaction features for students and professors.
- **Concurrent User Support**
  The system shall support at least **5,000 concurrent users** performing operations such as submitting feedback, viewing analytics, and accessing project information without noticeable performance degradation.
- **API Response Time**
  The login API shall respond within **2 seconds** for at least **95% of requests** under a load of 5,000 concurrent users.
- **User Action Responsiveness**
  The system shall respond to user-initiated actions (e.g., feedback submission, dashboard loading, analytics viewing) within **3 seconds** during normal operation.
- **Analytics Response Time**
  Analytics dashboards and graphical reports for professors shall load within **5 seconds**.
- **Fault Tolerance**
  The software shall incorporate basic fault-tolerance mechanisms to minimize service disruption and ensure continuity in the event of server or application-level failures.

4.2 Safety and Security Requirements

Authentication

- Access to the system shall be restricted to authenticated members of the institute, including students and professors.
- User authentication shall use the registered institutional email address as the primary identifier.
- A One-Time Password (OTP) shall be sent to the registered email address during login and password recovery.
- OTPs shall expire after **5 minutes**.
- Users shall create passwords with a minimum length of **8 characters**.
- Passwords shall be securely stored using hashing techniques.
- All authentication processes shall comply with industry-standard encryption protocols.

Access Control (Role-Based Access Control – RBAC)

The system shall enforce strict Role-Based Access Control to ensure proper segregation of responsibilities, such that:

- Students shall not have access to professor-specific analytics or administrative data.
- Professors shall not be able to modify or tamper with individual student feedback records.

CAPTCHA Verification

CAPTCHA verification shall be implemented during the login process to prevent automated access, brute-force attacks, and unauthorized login attempts.

Data Encryption

All sensitive data, including authentication credentials shall be encrypted during transmission using secure communication protocols.
Passwords shall be stored using salting and hashing methods, ensuring that even system administrators cannot view actual passwords.

Secure Data Storage

Feedback data, analytics outputs, and project-related information shall be securely stored to prevent unauthorized access, data breaches, or accidental data loss.

4.3 Software Quality Attributes

4.3.1 Usability

- **Intuitive Interface**

  The system shall provide a clean, intuitive, and user-friendly interface that allows users to navigate courses, feedback forms, dashboards, and campus interaction features with minimal effort.

- **Ease of Interaction**

  Feedback forms shall be simple, concise, and quick to complete, encouraging students to submit honest and consistent feedback after each lecture.

4.3.2 Availability

- **Monitoring and Failure Detection**

  The system shall continuously monitor server health and application logs to detect failures or crashes. Upon detection, the system shall automatically notify the system administrator to enable timely corrective action.
  In the event of a system failure, services shall be restored within **two hours** to minimize disruption to academic activities.

- **Scalability**

  The system shall be capable of scaling to handle increased usage during peak academic periods such as semester beginnings, examinations, and feedback submission deadlines.

- **Operational Accessibility**

  The platform shall remain accessible throughout the academic day with minimal downtime.

4.3.3 Reliability

- **Mean Time to Failure (MTTF)**

  The system shall achieve a Mean Time to Failure (MTTF) of **greater than one week** under normal operating conditions.

- **Testing Requirements**

  The software shall undergo comprehensive functional testing, load testing, and regression testing prior to deployment.

- **Data Accuracy and Consistency**

  Feedback analytics and project information shall be generated accurately and consistently to ensure reliable insights for faculty members.

- **Data Integrity**

  The system shall always preserve data integrity, including during periods of high user activity.

4.3.4 Portability

- **Platform Independence**

  The system shall be implemented as a web-based application and be accessible through all major modern web browsers.

- **Responsive Design**

  The application shall be responsive and function correctly across various screen sizes, including desktops, laptops, tablets, and mobile devices.

- **Browser Compatibility**

  The system shall minimize browser-specific dependencies to ensure consistent behavior across different platforms and environments.

5. Other Requirements

5.1 Authentication

Permission will be required for utilizing OTPs for the authentication process as this will require sending automated emails to the user's Email-Id.

Appendix A – Data Dictionary

A.1 User Database

Variable	Description	Requirements
User_ID	Unique identifier for each user	Unique for each user
Name	Full name of the user	String
Email	Institute email ID of the user	Valid institute email
Password	Encrypted login password	Encrypted string
Role	Role of the user	Student / Professor
Department	Academic department	String
Created_At	Account creation timestamp	Date-Time
Status	Account status	Active / Inactive

A.2 Student Database

Variable	Description	Requirements
Student_ID	Unique ID of the student	Unique
User_ID	Reference to User table	Must be valid User_ID
Roll_Number	Institute roll number	Unique
Branch	Academic branch	String
CPI	Cumulative Performance Index	Float (0–10)
Resume	Resume file uploaded by student	PDF/DOC format

A.3 Professor Database

Variable	Description	Requirements
Professor_ID	Unique ID of the professor	Unique
User_ID	Reference to User table	Must be valid User_ID
Designation	Academic designation	String
Department	Department of professor	String

A.4 Course Database

Variable	Description	Requirements
Course_ID	Unique identifier of course	Unique
Course_Name	Name of the course	String
Professor_ID	Instructor of the course	Valid Professor_ID
Semester	Academic semester	Integer
Credits	Course credit value	Integer

A.5 Enrollment Database

Variable	Description	Requirements
Enrollment_ID	Unique enrollment ID	Unique
Student_ID	Enrolled student	Valid Student_ID
Course_ID	Enrolled course	Valid Course_ID

A.6 Feedback Database

Variable	Description	Requirements
Feedback_ID	Unique feedback identifier	Unique
Student_ID	Student submitting feedback	Valid Student_ID
Course_ID	Course being reviewed	Valid Course_ID
Lecture_Date	Date of lecture	Date
Rating	Structured feedback rating	Integer (1–5)
Comments	Optional anonymous comment	Text
Submitted_At	Feedback submission time	Date-Time

A.7 Project Database

Variable	Description	Requirements
Project_ID	Unique project ID	Unique
Title	Project title	String
Description	Project description	Text
Professor_ID	Project posted by professor	Valid Professor_ID

Deadline	Application deadline	Date
Status	Project availability	Open / Closed

A.8 Project Application Database

Variable	Description	Requirements
Application_ID	Unique application ID	Unique
Project_ID	Applied project	Valid Project_ID
Student_ID	Applicant student	Valid Student_ID
CPI	Student CPI at time of application	Float
Branch	Student branch	String
Resume	Resume submitted	File
Applied_At	Application timestamp	Date-Time

A.9 Campus Interaction Database

Variable	Description	Requirements
Post_ID	Unique post identifier	Unique
User_ID	User creating the post	Valid User_ID
Content	Post or question content	Text
Is_Anonymous	Anonymous indicator	Boolean
Created_At	Post creation time	Date-Time

A.10 Reply Database

Variable	Description	Requirements
Reply_ID	Unique reply ID	Unique
Post_ID	Related post	Valid Post_ID
User_ID	User replying	Valid User_ID
Reply_Text	Reply content	Text
Created_At	Reply timestamp	Date-Time

A.11 Analytics Database

Variable	Description	Requirements
Analytics_ID	Unique analytics record ID	Unique
Course_ID	Course analyzed	Valid Course_ID
Average_Rating	Average feedback score	Float
Trend_Data	Historical feedback trend	JSON / Array
Generated_At	Analytics generation time	Date-Time

A.12 Authentication & Access Control Database

Variable	Description	Requirements
Auth_ID	Authentication record ID	Unique
User_ID	Associated user	Valid User_ID
Login_Time	Login timestamp	Date-Time
Logout_Time	Logout timestamp	Date-Time
Access_Token	Session authentication token	Encrypted

Appendix B – Group Log

S.No	Date	Timings	Venue	Description
1.	08/01/2026	17:00 to 19:00	RM Building	Several project ideas were brainstormed, including academic feedback systems, project collaboration portals, and campus interaction platforms. The Institute Academic Interaction & Feedback Platform was selected as the final project due to its relevance and scalability.
2.	09/01/2026	19:00 to 20:00	Hall 12	The project idea was finalized, the SRS template was studied, and tasks were distributed among team members.
3.	11/01/2026	19:00 to 20:00	Hall 12	Explored some more functionalities for the product and progressed with the SRS document.
4	13/01/2026	19:00 to 20:00	LHC	Explored various implementation tools and software for the project.
5.	14/01/2026	18:30 to 19:00	LHC	Team members discussed and shared updates on the SRS during the meeting.
6.	17/01/2026	19:00 to 19:30	Google Meet	Additional functionalities were explored, and progress was made on the SRS document.
7.	18/01/2026	10:00 to 12:30	RM Building	Progressed with SRS documentation work.
8.	18/01/2026	20:00 to 21:30	RM Building	Progressed with SRS documentation work.
9.	21/01/2026	19:00 to 20:30	Hall 12	The SRS document was finalized, typeset, and shared with the TA for final review.
10.	22/01/2026	19:00 to 21:30	Hall 12	It was reviewed for the final time, minute details were checked, and the document was prepared for final submission.